UNIVERSIDADE DE AVEIRO
DEPARTAMENTO DE ELECTRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA

Information and Coding (2026/27)

General info (version of September 9, 2026)

Instructors

- Armando J. Pinho (ap@ua.pt)
- António J. R. Neves (an@ua.pt)

Grading

- Theoretical component: Individual final written exam, 40%
- Practical component: Based on mini-project assignments, 60%
  - The assignments are performed in groups of three students
  - In the report of each assignment, it should be stated the percentage of participation of each member of the group
- Minimum of 8 points in each component

Learning objectives

At the end of this course, the students should know and understand:

- Fundamental concepts of digital signal processing
- The fundamentals of information theory and signal processing involved in efficient information representation
- The fundamentals and the need for the use of data compression algorithms
- The operation of the main data compression algorithms
- The relationship between data compression, the notion of algorithmic entropy and data analysis and exploration

Course contents

- Basic concepts of digital signal processing (sampling, quantization and representation of signals in the time and frequency domains)
- The notion of information: combinatorial, probabilistic, algorithmic
- Variable length codes
- Dictionary encoding
- Arithmetic coding

- Predictive, transform and sub-band coding
- Examples of some audio, image and video coding standards
- Application of compression techniques to data analysis and exploration

Website

- `http://elearning.ua.pt/`

Main bibliography

- Armando J. Pinho. *Some notes for the course "Information and Coding"*, Universidade de Aveiro, 2026
- Signals and Systems. Simon Haykin, Barry Van Veen. 2nd Edition, John Wiley & Sons, 2003
- Khalid Sayood. *Introduction to data compression*, Elsevier, 5th Ed, 2017
- David Salomon. *Data compression—The complete reference*, Springer, 4th Ed, 2007
- M. Li and P. Vitányi. *An introduction to Kolmogorov complexity and its applications*, Springer, 3rd Ed, 2008
- T. M. Cover and J. A. Thomas. *Elements Of Information Theory*, John Wiley & Sons, 2nd Ed, 2006

Note: Other editions of these textbooks are also appropriate.

Final remarks

- This is a 6 ECTS course, implying, on average, a total of about $6 \times 27 = 162$ hours of work.
- Academic dishonesty will not be tolerated. Academic dishonesty involves acts, such as
  - Cheating on an examination or quiz.
  - Substituting for another person during an examination or allowing such substitution for one's self.
  - Plagiarism. Act of appropriating passages from the work of another individual, either word for word or in substance, and representing them as one's own work.
  - Use of code written by other may be accepted, but always with the due reference to the authors. In case of doubt, ask the instructors.
  - Collusion with another person in the preparation or editing of assignments submitted for credit, unless such collaboration has been approved in advance by the instructors.
- Use of AI tools
  - The use of AI tools is permitted (and encouraged) to assist in the development of the practical assignments. However, students will be frequently asked by the instructors to explain the proposed solutions. Failing to do so in a satisfactory way implies a low or even zero mark.
  - The use of AI tools is not permitted during the written exams.
- If you have doubts regarding a certain action, ask the instructors.